Credit: IIT Bombay/NIT Srinagar/IUST

The Ruhdaar ventilator

nised compression, along the lines of the initial designs explored by the IIT Kanpur team. The innovative design mechanises the part that is manual, and allows the doctor to work at a distance from the patient. Dr. Arpan Gupta, Associate Professor, School of Engineering, IIT Mandi, said "Designed especially for COVID-19 pandemic, this low-cost ventilator can be operated manually as well as using a smartphone app connected over Wifi. It can offer some protection to our medical staff, who can operate the ventilator remotely." Aerobiosys Innovations, a startup supported by IIT-Hyderabad developed a low cost version of a ventilator that they were already making, to address the demand of the pandemic.

The real world problem of providing sufficient ventilators for India was completely met by the local manufacturing industry in time. In fact, the lack of cess or duties on imports, a ban on exports that was not lifted for too long, and boosted manufacturing has now led to an oversupply of ventilators. At the same time, hesitancy in adoption of the new local ventilators means that there is paradoxically, a simultaneous shortage of the ventilators.

RESPIRATORS

At the outbreak, India needed quality respirators that could provide adequate protection against the pandemic, and there was a shortage of these. E-spin nanotech, another one of the startups incubated at IIT Kanpur began to use its tiny electron spinning machines (that it developed and manufactures) to produce respirators on campus, in an automated production process in June 2020. These masks were branded as "Swasa", and over the months the production was ramped up through manufacturing partners, and distribution was also scaled through partnerships with ecommerce portals such as Flipkart. The expertise of the startup was in textiles.

Nanoclean, a Delhi-based startup that had already won awards for its innovative products to tackle the pollution, developed and began to produce respirators after receiving grants from the government and IIT Delhi. Nanoclean has been steadily increasing production capacity since the onset of the pandemic. Nirvana Being, another company founded in Delhi and focused on combating pollution, also used nanotechnology in the production of its N95 masks. E-Tex, an IIT-Delhi startup with expertise in smart textiles started producing its Kawach respirator, which was available for as little as ₹45, boosted the production, and began providing antiviral protection kits as well. Check out their website if you want to pick up some antimicrobial tshirts. All these masks are made up of layers of material, with each layer specialising in filtering large particles, small particles, having antiviral or antibacterial properties or directing away condensation. Most importantly, all of these masks are as effective at keeping the virus in as well as keeping the virus out, which is the need of the hour of the pandemic.

N95 is the American standard used by the Centers for Disease Control and Prevention (commonly known as the CDC) and certified by National Institute for Occupational Safety & Health (NIOSH), whose website has a list of all N95 manufacturers. There are other similar or comparable standards, such as the EN 149 standard in Europe with the FFP2 designation for masks that protect against very fine particles and filter out 94 percent of the aerosols, and the FFP3 designation for masks that filter out 99 percent of particles. Australia has the AS 4381:2015 standard with masks graded on three levels. India too has its own standard, known as IS 16289:2014, which also grades masks on three levels. America has another certification standard by the ASTM F 2101 : 2014, American Society for Testing and Materials for bacterial filtration efficiency, not just aerosols. One or more of these numbers may be printed on the masks.

The procedures for getting the products certified and licensed for production in India were relaxed during the pandemic, which does not mean the grading itself was relaxed. Instead the government gave certain allowances for companies that did not have the facilities needed for in house testing of the products. Despite all of this, the most well known certification is N95, and the manufacturers of the masks are forced to use the N95 certification in their promotional material because of the high demand by the consumer. For example, a mask that is FFP2 certified, may need to call itself the N95 mask so potential buyers can understand that it is the same thing.

The problem goes back to the SARS outbreak in South Asia when the WHO initially recommended only N95 masks to deal with the pandemic. Since then, the market has been flooded with cheap and dubious masks, sporting bogus certifications, some of which are patently ridiculous (such as N999). Before the nationwide lockdown itself, there were vendors outside railway stations hawking masks of spurious quality.

Prateek Sharma, CEO and Director of Nanoclean Global Pvt Limited says, "as the demand surged in the Indian market from the mid of March 2020, there was a scarcity of N95 grade face mask. I would say we have seen every possible form of malpractices in the facemask business during this time. Infiltration and manufacturing of fake masks which are ineffective , profiteering by few mask manufacturers, fake copy of branded products, repackaging of used masks – our country has seen it all. " Nanoclean used their expertise in making products to protect